

SODANKYLA VUOTSO, Finland

WMO: 028160

Lat: 68.084N

Lon: 27.185E

Elev: 816

StdP: 14.27

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.6	-22.8	-34.9	0.8	-28.4	-28.5	1.1	-22.6	14.3	23.8	12.7	24.0	2.2	340	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	76.8	60.5	72.1	59.0	68.2	57.6	63.1	72.0	60.9	69.3	59.0	65.9	6.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	79.7	65.0	57.8	73.5	63.0	55.9	68.7	61.3	29.0	71.9	27.4	69.6	26.1	66.2	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	11.0	9.7	-36.0	80.5	3.5	4.9	-38.5	84.0	-40.5	86.8	-42.4	89.5	-45.0	93.1
			-35.4	65.6	4.3	3.3	-38.4	68.0	-40.9	69.9	-43.3	71.7	-46.4	74.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.5	6.1	8.8	16.0	28.3	40.7	49.8	56.4	51.7	43.8	30.2	19.3	13.3	(6)
(7)		DBStd	19.78	13.45	14.79	11.29	8.56	8.14	7.68	7.19	6.66	6.33	8.34	11.21	13.02	(7)
(8)		HDD50	7575	1360	1153	1055	650	307	96	21	60	201	614	920	1138	(8)
(9)		HDD65	12610	1825	1573	1520	1100	753	459	278	415	635	1079	1370	1603	(9)
(10)		CDD50	459	0	0	0	0	19	91	220	113	16	0	0	0	(10)
(11)		CDD65	18	0	0	0	0	0	3	12	3	0	0	0	0	(11)
(12)		CDH74	247	0	0	0	0	8	38	167	34	0	0	0	0	(12)
(13)		CDH80	37	0	0	0	0	0	3	27	7	0	0	0	0	(13)
(14)	Wind	WSAvg	4.7	4.1	4.6	5.1	5.0	5.3	5.5	4.8	4.6	4.6	4.2	4.3	4.5	(14)
(15)	Precipitation	PrecAvg	20.5	1.2	1.2	1.1	1.3	1.5	2.5	2.6	1.9	2.0	1.7	1.3	(15)	
(16)		PrecMax	26.6	2.9	3.1	2.4	3.9	3.3	4.3	4.8	5.2	3.7	4.1	2.8	2.5	(16)
(17)		PrecMin	15.4	0.4	0.4	0.2	0.4	0.4	0.4	0.7	0.6	0.3	0.4	0.3	0.2	(17)
(18)		PrecStd	3.4	0.6	0.7	0.6	0.7	0.6	1.0	1.0	1.1	0.9	1.1	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	36.4	39.9	52.4	73.4	79.1	83.8	78.7	63.8	47.6	39.9	35.7	(19)
MCWB			30.9	34.6	34.7	43.8	55.7	62.0	63.7	63.9	55.3	44.7	39.2	34.6	(20)	
2%		DB	29.3	32.7	36.4	45.5	65.8	72.5	78.7	71.8	60.1	45.7	35.3	33.4	(21)	
		MCWB	28.8	31.6	32.4	38.3	52.0	58.2	60.9	60.3	53.2	43.8	34.9	32.7	(22)	
5%		DB	27.1	30.9	33.6	42.2	59.6	67.4	74.3	67.4	57.4	43.3	33.6	31.9	(23)	
		MCWB	26.7	29.9	31.2	36.3	48.7	55.3	59.6	58.4	51.7	41.8	33.3	31.6	(24)	
10%		DB	23.9	28.2	31.6	39.4	54.3	63.1	70.0	63.4	54.5	40.9	32.3	29.8	(25)	
		MCWB	23.4	27.4	29.7	34.3	46.4	53.1	58.5	56.4	50.5	39.9	32.1	29.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.2	35.1	35.5	43.7	56.5	63.8	67.1	65.3	57.4	46.5	39.6	34.9	(27)
MCD B			31.7	36.5	39.2	51.8	71.0	75.5	74.2	75.7	61.8	46.6	39.6	35.5	(28)	
2%		WB	28.8	32.1	33.4	39.1	53.4	60.9	64.0	61.7	54.8	44.0	34.9	32.9	(29)	
		MCD B	29.2	32.6	35.5	44.4	62.9	70.9	72.7	68.8	58.0	45.3	35.2	33.3	(30)	
5%		WB	26.7	29.9	31.5	36.9	49.8	57.5	61.8	59.3	52.8	41.9	33.3	31.6	(31)	
		MCD B	27.1	30.7	33.2	41.1	57.5	63.6	70.7	65.6	56.0	43.2	33.5	32.0	(32)	
10%		WB	23.4	27.5	29.8	35.4	46.8	54.5	60.1	57.5	50.9	40.0	32.2	29.3	(33)	
		MCD B	23.8	28.2	31.6	38.6	53.6	61.7	68.0	61.9	53.8	40.9	32.3	29.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.8	15.5	20.3	17.2	15.8	15.7	16.6	16.5	13.9	9.8	12.2	14.7	(35)
MCD BR			15.0	11.4	14.2	18.0	25.1	22.6	25.1	23.2	20.2	10.5	8.0	9.3	(36)	
MCWBR			15.1	10.6	11.8	11.8	13.4	10.4	10.5	12.6	13.3	8.9	8.1	9.3	(37)	
5% WB		MCD BR	14.8	11.0	12.0	16.4	23.0	19.0	19.3	19.5	15.5	9.4	7.9	9.3	(38)	
		MCWBR	14.9	10.5	10.4	11.0	13.1	9.9	9.8	11.5	11.1	8.3	8.1	9.4	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.190	0.249	0.287	0.318	0.337	0.339	0.351	0.333	0.305	0.267	N/A	N/A	(40)	
taud		1.916	2.080	2.159	2.209	2.400	2.480	2.465	2.491	2.545	2.301	N/A	N/A	(41)		
Ebn at Noon		31	172	235	262	272	274	266	259	239	170	N/A	0	(42)		
Edh at Noon		3	15	25	33	30	29	29	25	18	12	N/A	0	(43)		
(44)	All-Sky Solar Radiation	RadAvg	12	159	634	1222	1476	1573	1436	1035	541	177	29	0	(44)	
(45)		RadStd	1	17	42	75	143	134	154	95	53	31	4	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page